

Project Design Phase-II - Technology Stack

Date	Team ID	Project Name	Maximum Marks
18 June 2026		ShopEZ E-Commerce Platform	4 Marks

1. Technical Architecture Guidelines

Our e-commerce solution is built on a clean MERN Stack model separated into client and server components:

- **Frontend (Single Page Application):** Vite/React, custom Context-driven state managers, responsive HSL-based styles.
- **Backend (RESTful Express API):** Express routing, body-parser middleware, JWT headers.
- **Database (NoSQL):** Mongoose schemas mapping database structures, localized index parameters.

2. Components & Technologies

Layer	Component	Chosen Technology	Reason for Selection
Frontend UI	Client Interface	React + Vite	Fast component re-rendering, hot-reloading.
	State Manager	React Context API	Avoids prop drilling, handles Cart/Auth states.
App Logic	Server-Side Core	Node.js + Express	Single language stack, non-blocking asynchronous event loop.
Database	Database Engine	MongoDB	Document store maps naturally to JSON objects.
Auth	Security Tokens	JWT (jsonwebtoken)	Stateless session management.
Encryption	Passwords	bcryptjs	Secure cryptographic one-way salting and hashing.
Networking	API Calls	Axios	Automatic JSON parsing and request/response interceptors.

3. Application Characteristics

- **Open-Source Frameworks:** Fully utilizes npm open-source ecosystem libraries.
- **Security:** JWT verification on routes `/api/users/profile`, `/api/orders`, `/api/admin`.
- **Performance:** Client caches static product items in state; server connects with optimized database cursor indices.